STATS 210 **Team Task 1 : Conditional Probability**

Summarise your team answers on ONE piece of paper. You do not need to hand it in.

Marks are awarded for attendance and participation. Marks written below are for guidance only. Use these sessions to practise your creativity and ingenuity!

Mr Tambourine runs a musical coffee shop called Cafe Swan Cake, that attracts musical customers from all over the world. The sample space is $\Omega = \{\text{customers}\}$.

Define events $I = \{\text{customer is Irish}\}$; $B = \{\text{customer plays the Banjo}\}$.

(a) Write down **four different sentences** that express the information $\mathbb{P}(B \mid I) = 0.4$. Each sentence should have a different wording or sentence structure, but each sentence should unambiguously convey that $\mathbb{P}(B \mid I) = 0.4$. The sentences should be written in natural English, using the words 'Irish' and 'banjo' rather than the letters I and B. (4)

(b) Write down **two** different sentences that express the information $\mathbb{P}(B \cap I) = 0.2$. (2)

(c) Draw a diagram, showing the sample space Ω, and two possible events B and I, that clearly conveys the two pieces of information that $\mathbb{P}(B \cap I) = 0.2$ and $\mathbb{P}(B \mid I) = 0.4$. Use **areas** on the diagram to convey probabilities, and label your diagram so that it clearly depicts both the required quantities and the correct areas. [**Hint: rectangular regions on a diagram will convey areas better than circular regions.**] (4)

(d) Draw two further diagrams showing Ω, B, and I, where $\mathbb{P}(B \cap I) = 0.2$ and $\mathbb{P}(B \mid I) = 0.4$.

(i) The first diagram should show B and I as **independent events.** Include any brief pieces of working or calculation that have helped you to create this diagram.

(ii) The second diagram should show B and I as **non-independent events.** (6)

(e) Suppose that B and I are **independent** events. You want to explain to a high-school student, in words, why independence means that $\mathbb{P}(B \cap I) = \mathbb{P}(B)\mathbb{P}(I)$. Write down your **brief** explanation, in **at most two sentences**. If you wish, you may use numeric values calculated earlier in the question as part of your explanation. (4)

If you have extra time: Draw a diagram to explain why, if events $I_1, \ldots, I_n$ are mutually exclusive and $\mathbb{P}(B \mid I_r) = k$ (a constant) for all $r = 1, \ldots, n$, then $\mathbb{P}(B \mid I_1 \cup I_2 \cup \ldots \cup I_n) = k$ also. *This is an example of a question that is easily solved visually, once you understand visual representations of conditional probability, but is tedious to solve algebraically.*

BEFORE YOU LEAVE: speak to a tutor, get copies of the answer sheet, and discuss how your answers agree with the suggested answers.

Note: Venn diagrams are poor tools for conveying the concept of independence, though it is possible as you have just shown. Hopefully part (d) will have convinced you that it is quite tricky to depict independence on a diagram. Someone looking at a Venn diagram would not usually be able to see-at-a-glance whether or not the events are independent.